Jade Clement

UofT Engineering Science Student

Ambitious and enthusiastic student eager to gain experience and skills required to make an impact on real world applications of engineering

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EDUCATION

Engineering Science: Machine Intelligence

University of Toronto

09/2020 - 05/2026

Relevant Courses

- MIE429: Machine Intelligence Capstone Design
- ECE324: Machine Intelligence, Software and Neural Networks
- CSC413: Neural Networks and Deep Learning
- ECE353: Systems Software
- ECE358: Foundations of Algorithms (Data Structures)
- MIE424: Optimization in Machine Learning
- ROB311: Artificial Intelligence
- ECE421: Introduction to Machine Learning
- MIE451: Decision Support Systems
- ECE368: Probabilistic Reasoning
- CSC343: Introduction to Databases

PROJECTS

Full Stack Web Application (08/2025-09/2025)

- Designed and built a club management platform serving **60+ active users** using **React.js**,**Node.js/Express** and **PostgreSQL.**
- Integrated **AWS S3** and **Cloudfront** for scalable image storage and persistence data management.
- Developed real-time workout signup and waitlist system with automated email notifications.
- Built an admin dashboard for member management, attendance tracking and analytics.

ML Document Classification (10/2023)

- Implemented classifiers with different encodings (binary, tf-idf) using **scikit-learn, numpy, pandas, and scipy libraries**
- Varied the feature set, feature encoding, and hyperparameters to optimize performance on classifying 20,000 newsgroup documents

Capstone Project (08/2023 - 12/2023)

- Collaborated with teammates and clients at Perimeter AI to brainstorm solutions to their data imbalance problem and ultimately help them improve their AI algorithm that detects cancer on OCT scans
- Explored techniques such as CGANs, Object Detection and Elastic Transform to generate new data leading to a **5% testing accuracy increase on ResNet-18**

Natural Language Projects (10/2023 - 12/2023)

- Performed **sentiment analysis** on hotel review data scraped from Tripadvisor and plotted output to visualize results
- Designed a **prompt engineering** experiment to evaluate various prompt styles for a pretrained LLM on a task involving extracting specific information from email chains

Web Development (06/2023 - 08/2023)

- Programmed and designed my personal website using HTML, CSS and Javascript which can be found at jadeclement.com

Self-balancing Robot (01/2022 - 04/2022)

- Brainstormed, planned, designed and built a self-balancing robot prototype with 4 team members

SKILLS

Python | ML/DL/RL | MatLab | C | HTML | CSS | JavaScript | PyTorch | Tensorflow | Keras | Git | Data Structures | Communication | PostgreSQL | Django | Node.js

EXPERIENCE

Software Engineering Intern

SingleKey — Toronto, ON

05/2024 - 04/2025

- Developed backend features using **Django** and **PostgreSQL**
- Built **full-stack** features using **Python** for backend and **Node.js** for frontend
- Contributed to a robust and maintainable codebase through active participation in code reviews
- Researched emerging technologies to support future platform enhancements, including AI initiatives

Machine Learning Research Assistant

Robotics and Control Lab — Vancouver, BC

05/2023 - 08/2023

- Reviewed academic papers on generative models, especially **diffusion models**
- Designed and planned a machine learning model to generate echocardiogram videos by collaborating with team members
- Programmed and debugged in Python within Docker containers to reproduce results of the paper in the same field

Medical Imaging Research Assistant

UBC Centre for Heart & Lung Innovation — Vancouver, BC

05/2021 - 08/2021

- Delivered a presentation to a large audience of research professionals and doctors
- Analyzed 50 cardiac CT cases for an inquiry for a cardiovascular diseases biomarker from Lung CT analysis
- Resolved technical issues of the image recognition software that creates 3D models of the lungs; troubleshooting
- Concluded that quantitative lung analysis is feasible within Cardiac CT field of view

AWARDS

Qualifier for the Worlds Triathlon Championship (2024 & 2026)

Dean's Merit Award from the University of Toronto (09/2020)

LANGUAGES

English ●●●●●

French ●●●●○

Spanish ●○○○○